

- Change to the tab **Statistics Values**. Type in the parameters of the distribution for which you want to investigate the effects of the bounds. Then, click **Start** to compute the real mean value and the real standard deviation of the distribution with bounds.

After the calculation is done, you can click **Show** to view the density function and the bounds. The density visualizes how the bounds cause the mean value to shift.

The solid vertical line designates the real mean value of the distribution with bounds.

The dotted line designates the mean value of the distribution without bounds.

In addition, the *TruncDistribution* show the probability of a hit, with which a random number of the distribution without bounds falls in between the bounds, which you typed in.

Plant Simulation rolls the dice until a random number falls in between the bounds of the distribution with bounds. When its probability is too small, it may happen that the **maximum number of samples**, which you typed in under **File > Preferences > Simulation** does not suffice and *Plant Simulation* interrupts the simulation.

- On the tab **Parameterization** you determine the parameters of the distribution, so that the distribution with bounds arrives as the mean value, which you set.

To a certain degree you execute the reverse conversion of the calculation task you defined on the tab **Statistics Values**.

Click **Start** to start computing. Not each mean value between the bounds is viable.

Back to [Step 3: Decide which Distribution to Use](#)

Running Simulation Experiments

Run Simulation Experiments

After creating your model, you have to validate it, i.e., you have to make sure that the model displays the intended properties and shows the intended behavior. In a simulation study you will consider how certain model parameters as input values influence result values as output values.

You might, for example, execute a number of simulation experiments with the [ExperimentManager](#)  to:

Procedure

1. Arrive at **statistically safe results**.

When you re-created an existing plant, you will compare the data you collected on the shop floor with the results of the simulation run.

When your model contains random components, you will have to investigate how reliable the data resulting from the simulation runs is. This is especially important for data you use for drawing conclusions about the plant you modeled. Make sure to assign different seed values to the different random components of the random number streams. To consider how the random components

statistically influence the results the *ExperimentManager* will change the seeds of the corresponding random number streams for each set of input values. This way you arrive at a set of observations for the output values under consideration.

For this reason the *ExperimentManager* generates a statistical evaluation for each set of input values you enter. More detailed, for each output value and for fixed input values you will receive a confidence interval and a min-max interval combined in a box plot with a desired level of confidence.

A confidence interval, which results from the observations, is an interval of numbers that contains the true mean value of the random number with the confidence level which you entered. The mean value of a random component of the simulation model is unknown, and it cannot be measured exactly, as the simulation runs only provide a finite number of realizations of the corresponding random output values.

To statistically hedge the resulting data for random data, you not only average out the results of the different simulation experiments, but you also have to decide if the data permits to draw conclusions about the simulated system, despite fluctuations of the resulting data.

Be aware that one cannot draw conclusions with absolute safety. One can only make statements about a system with random components with a confidence level (level of significance) you define. Compare the chapter about [confidence intervals](#) in the documentation that comes with the library *DataFit*.

2. Optimize the results of the simulation experiment.

To determine the parameters that will lead to the desired simulation results, or to optimize the results of the simulation experiment, you can also use [Genetic Algorithms](#).

Note

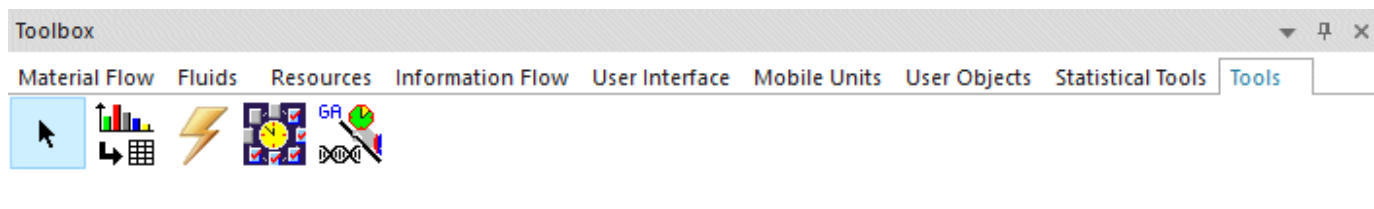
We modeled the dialog of the *ExperimentManager* with the object *Dialog*. For this reason the **dialog box** of the *ExperimentManager* does not use the Siemens PLM standard design, but the Windows design.

Go to [Run Experiments with the ExperimentManager](#)

Running Experiments with the ExperimentManager

Run Experiments with the ExperimentManager

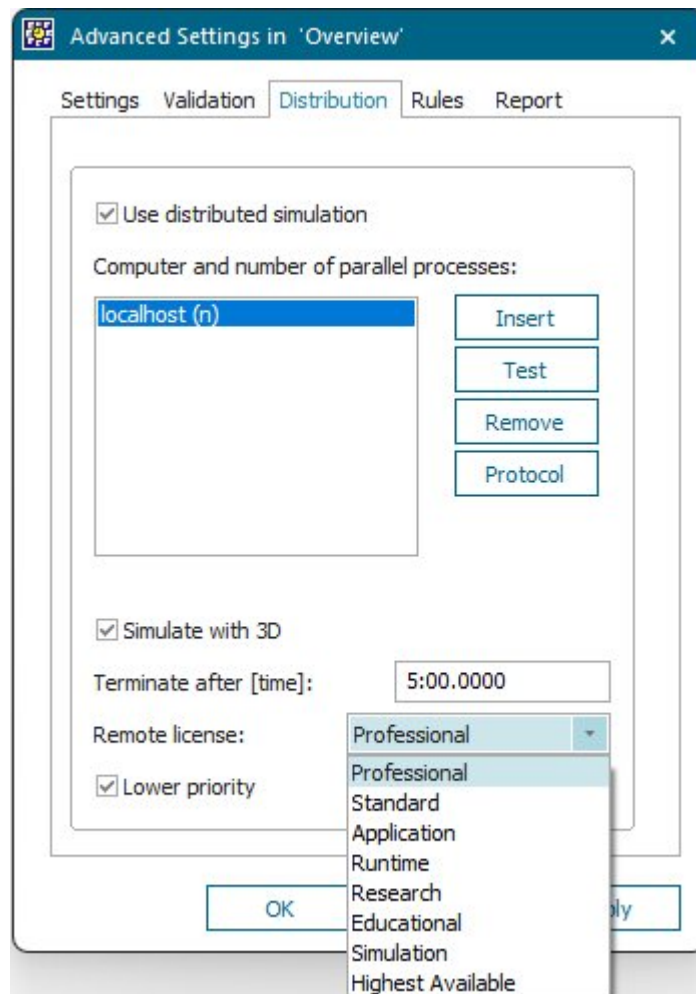
You find the *ExperimentManager* in the *Tools* folder in the *Class Library* or on the toolbar **Tools** in the *Toolbox*.



Your *Plant Simulation* license determines how you can use the [ExperimentManager](#) .

- With a **Runtime license** you can execute experiments with a simulation model. You can modify parameters in tables. You cannot change the structure of the *ExperimentManager* model.
- With an **Application license**, you can also change the structure of the model.
- With a **Professional license**, a **Standard license**, or a **Foundation license** you can, in addition, define the experiments in a *Method*.
- The **Runtime license** and the **Student license** cannot control distributed simulations.

You can use computers with the following license types as simulation machines:



With the setting **Highest Available** for the remote license you can use the different license types for the simulation machines. This way you can also use the controlling computer as a simulation machine.

Note

A **simulation study** contains several **experiments**. Each **experiment** executes several **simulation runs**, each of which leads to an observation.

Running a Simple Simulation Study

Run a Simple Simulation Study

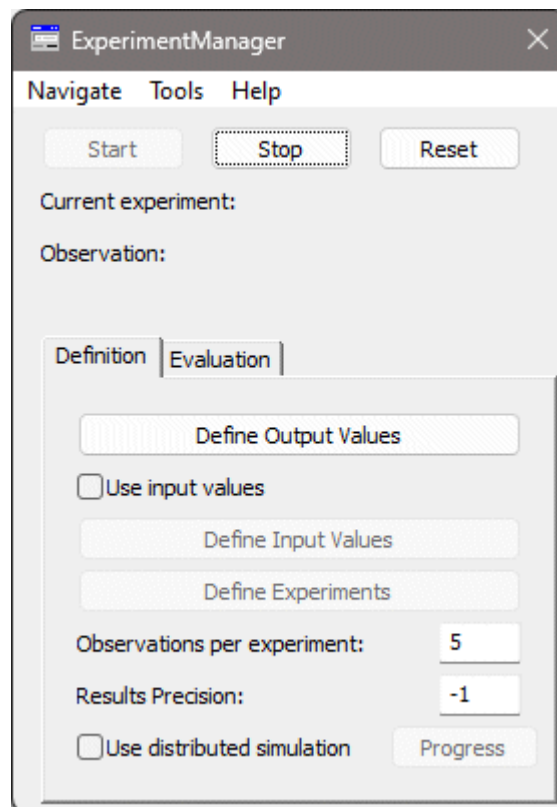
You can execute a simple simulation study in three quick-and-easy steps.

- **Step 1: Define input and output values of the experiments.**
- **Step 2: Run the experiments with the defined settings.**
- **Step 3: Evaluate the results of the simulation study.**

You can also [Refine the Settings](#) and [Set Static Parameters](#) and [Set Dynamic Parameters](#).

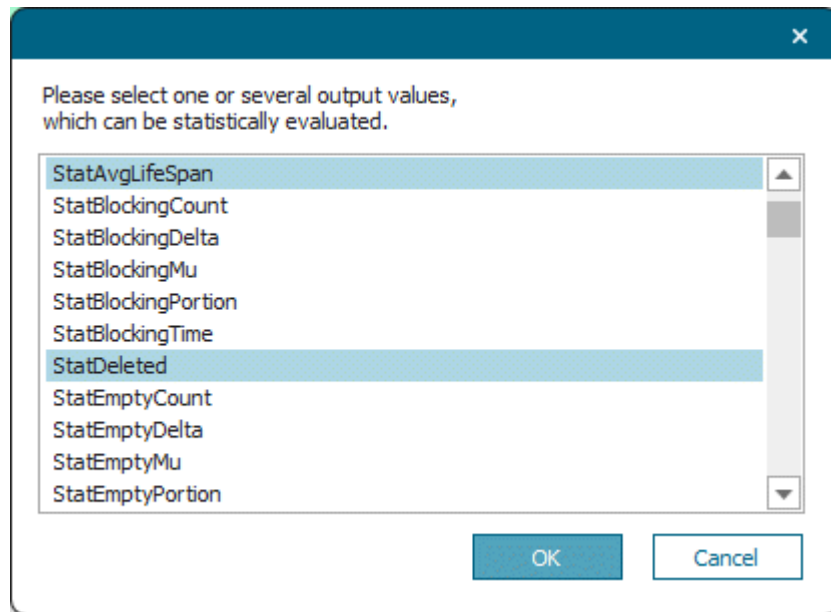
Step 1: Define input and output values of the experiments

In a simulation study you'll examine how a set of input values, which are the model parameters, affects output values. The output values can be observed after a single simulation run. Each simulation study needs at least one output value.



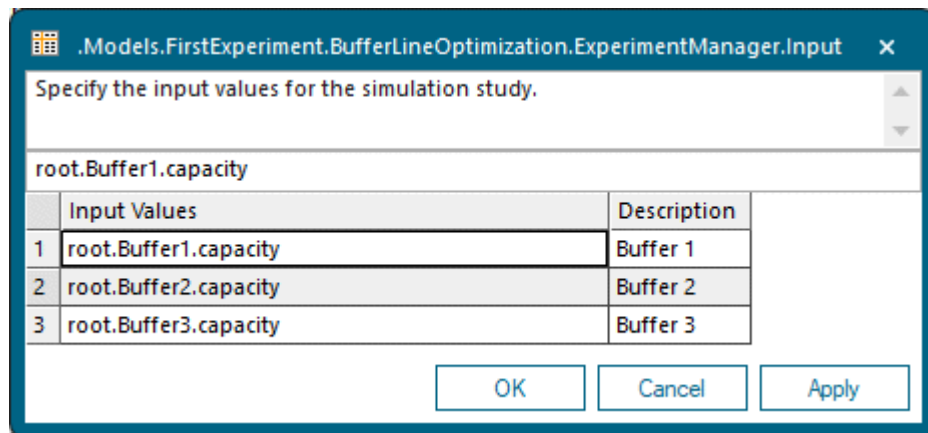
To open the table into which you enter the output values, click the button **Define Output Values** on the tab **Definition**. An output value can be a *method* or an *attribute* of an object in your model. You can enter a description for the output values which the *ExperimentManager* show in all evaluations. This makes the results of the simulation study easier to understand.

When you drag an object onto the *ExperimentManager*, it offers all numerical attributes and methods which can be used as output values.



If the results of a simulation run are provided in a *Plant Simulation* table, you can use the entire table as output value of the study. To designate a table as the results table drag it onto the *ExperimentManager*.

To open the table into which you enter the input values, click the button **Define Input Variables** on the tab **Definition**. An input value can be an attribute of an object or an entry in a table. Before executing an experiment, the *ExperimentManager* sets the input values in the simulation model. These input values are not changed in the simulations runs of an experiment.



Hold down the **Shift** key and drag an object onto the *ExperimentManager* and drop it there. This shows all attributes of the object in a dialog. To select one or more attributes, hold down **Ctrl** and click the attributes you are interested in. This opens the data table of all input values. Then, add other values. You can also enter a description of the input value which the *ExperimentManager* uses in all evaluations. The *ExperimentManager* checks the entries by closing this data table.

Enter the input values for each experiment.
If you want to run single experiments and preserve already simulated experiments, mark the experiments as active.

true

	Name	Active	Buffer 1	Buffer 2	Buffer 3
1		true	2	2	2
2		true	2	2	5
3		true	2	2	8
4		true	2	5	2
5		true	2	5	5
6		true	2	5	8

OK Cancel Apply

Input values can also be parameters of probability distributions, which you have to type into the table. The *ExperimentManager* recognizes these probability distributions when holding down **Shift**, dragging an object onto the *ExperimentManager* and dropping it there: *Uniform*, *Triangle*, *Normal*, *Lognorm*, *Erlang*, *Negexp*, *Geom*, *Hypgeo*, *Weibull*, *Binomial*, *Poisson*, and *Beta*. To select one or several attributes of interest, hold down **Ctrl** and click the attributes of interest.

Enter the input values for each experiment.
If you want to run single experiments and preserve already simulated experiments, mark the experiments as active.

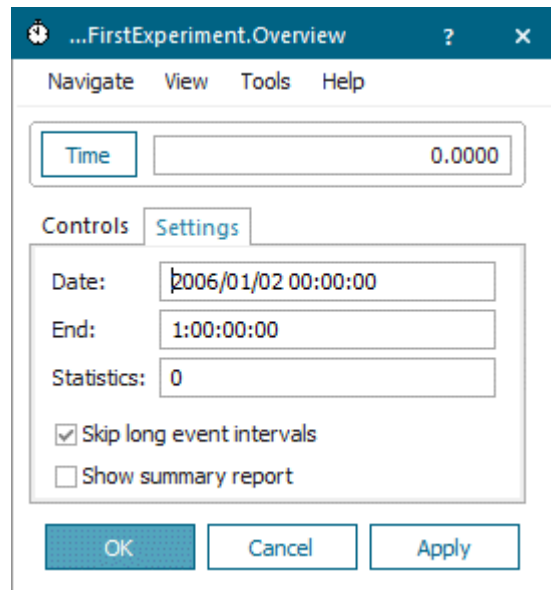
true

	Experiments	Active	Interarrival time	Intermediate store size	Availability of the tool
1		true	20.0000	2	50
2		true	20.0000	2	90
3		true	20.0000	4	50
4		true	20.0000	4	90
5		true	20.0000	6	50
6		true	20.0000	6	90
7		true	20.0000	8	50
8		true	20.0000	8	90
9		true	20.0000	10	50
10	Best results	true	20.0000	10	90

OK Cancel Apply

To set the input values for the experiments, as shown in the figure below, click the button **Define Experiments** on the tab **Definition**.

After you have finished creating your simulation model, you can start running experiments with it. A simulation run always has to end. Many models might continue running after the processes, which interest you, have been simulated, for example to simulate failures, etc. When you use the method *stop* of the *EventController* in your simulation model, you have to call the method *endSim* of the *ExperimentManager* at the appropriate place. Instead, you can also set the duration of the simulation, by either entering an **End** time in the *EventController* or setting it with the attribute *End*.



See also

[Select the Output Values for the Experiments](#) in the *Reference* part of the *Help*

[Select the Input Values for the Experiments](#) in the *Reference* part of the *Help*

[Define the Input Values in the Experiment Table](#) in the *Reference* part of the *Help*

Video on YouTube

<https://youtu.be/HKDhTbohV6M?si=tcZ63Doub1SXGZw0&t=68>

<https://youtu.be/HKDhTbohV6M?si=1UaMMz81vspoflcj&t=168>

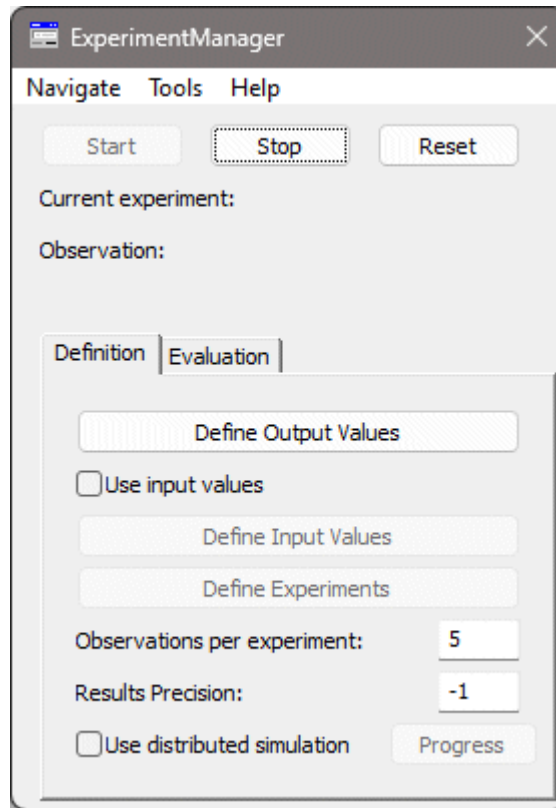
Step 2: Run the experiments with the defined settings

In this step run the experiments with the defined settings.

Proceed as follows:

Procedure

1. Click **Reset** to reset the *ExperimentManager*.
2. Click **Start** to start the **experiment run**.



Depending on the size of your simulation model, running the experiments can take a while. *Plant Simulation* shows the currently running simulation in the dialog. At the end of the experiment run the *ExperimentManager* opens a report.



When you defined several experiments with many simulation runs that take a long time to be executed, you can run them overnight.

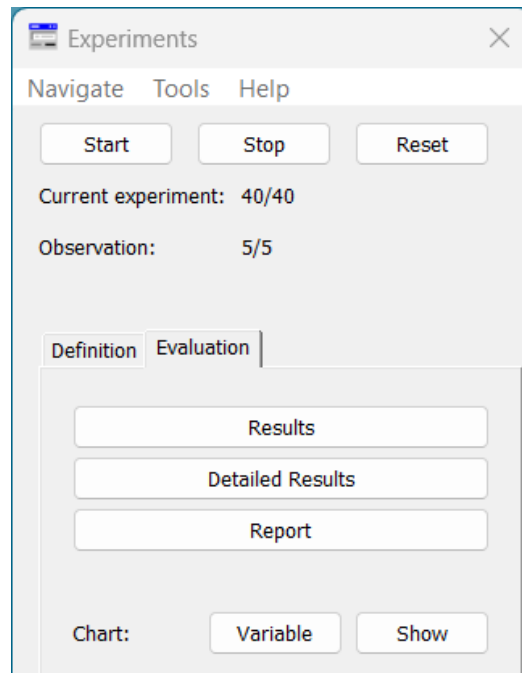
See also

[Controlling the Experiment Runs](#) in the *Reference* part of the *Help*

Step 3 Evaluate the results

Step 3: Evaluate the results of the simulation study

After the *ExperimentManager* has executed the simulation experiments, you will want to examine the results.



You can:

- [View the Results as a Table.](#)
- [View the Results in a Chart.](#)
- [View the Results in a HtmlReport.](#)

See also

[Viewing Evaluations](#) in the *Reference* part of the *Help*

View the Results as a Table

In this step we display the results of the simulation experiments in the result tables.

Proceed as follows:

Procedure

1. Click **Open Results** on the tab **Evaluation** to show an overview for all simulation experiments. The table *Results* show all input values and all mean values of the output values. To show a results table in a subtable, select the respective entry in a cell on the table and press **F2**.

	Experiment	Buffer 1	Buffer 2	Buffer 3	Throughput no failure	Throughput with failure	Throughput with Buffers
1	Exp 01	2	2	2	477	349.8	387.6
2	Exp 02	2	2	5	477	349.8	396.4
3	Exp 03	2	2	8	477	349.8	399.6
4	Exp 04	2	5	0	477	349.8	406.8
5	Exp 05	2	5	5	477	349.8	403.6
6	Exp 06	2	5	8	477	349.8	406.8
7	Exp 07	2	8	0	477	349.8	409.6

Note

You can execute an experiment run without any input values. But each simulation study requires at least one output value.

2. Click **Detailed Results** to show the values for all output values.

To open a subtable, select the cell and press F2.										
	Experiment	Throughput no failure	Observations	Throughput with failure	Standard Deviation	Minimum	Maximum	Left interval bound	Right interval bound	Observations
1	Exp 01	477	Table 01	349.8	5.21536192416044	346	359	343.324272861126	356.275727138874	Table 01
2	Exp 02	477	Table 02	349.8	5.21536192416044	346	359	343.324272861126	356.275727138874	Table 02
3	Exp 03	477	Table 03	349.8	5.21536192416044	346	359	343.324272861126	356.275727138874	Table 03
4	Exp 04	477	Table 04	349.8	5.21536192416044	346	359	343.324272861126	356.275727138874	Table 04
5	Exp 05	477	Table 05	349.8	5.21536192416044	346	359	343.324272861126	356.275727138874	Table 05
6	Exp 06	477	Table 06	349.8	5.21536192416044	346	359	343.324272861126	356.275727138874	Table 06
7	Exp 07	477	Table 07	349.8	5.21536192416044	346	359	343.324272861126	356.275727138874	Table 07

The observations of each experiment represent a sample for each output value from which these are computed:

- Mean value
- Standard deviation
- Minimum
- Maximum
- Left bound of the confidence interval
- Right bound of the confidence interval

The confidence intervals belong to the **Confidence interval** which you entered under **Tools > Advanced Settings > Settings**.

If you defined a table as output value, the table *DetailedResults* provides subtables with the following statistical values of the numerical entries of this table:

- Mean value
- Standard deviation
- Minimum
- Maximum

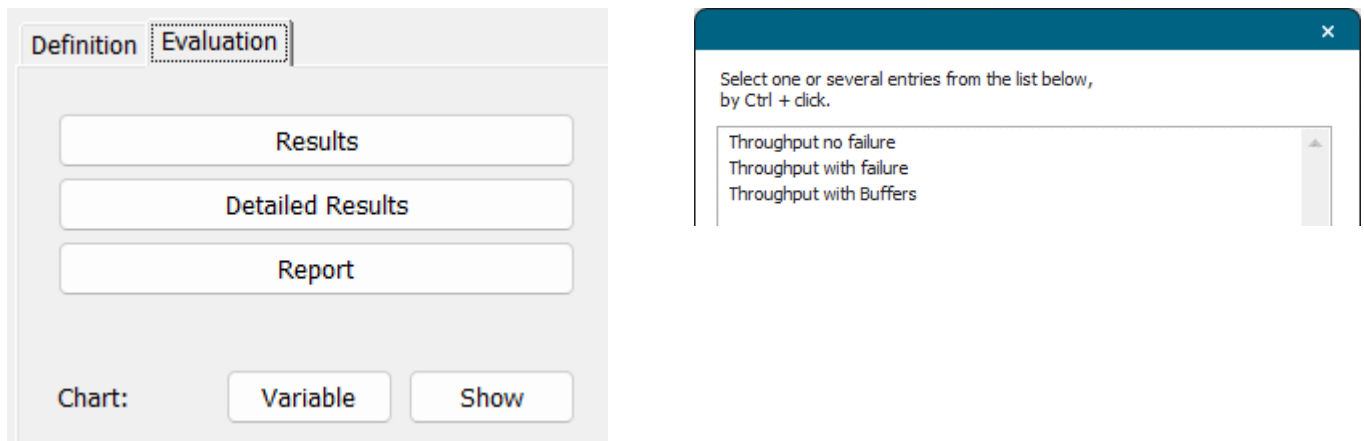
The column **Observations** provides the results of all simulation runs for all output values.

See also

[View Output Values of Interest](#) in the *Reference* part of the *Help*

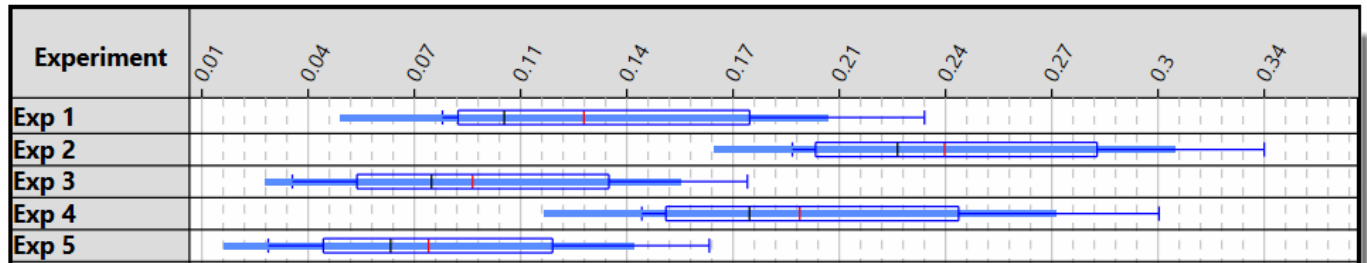
View the Results in a Chart

The *ExperimentManager* can generate a chart of the selected output values.



You can select to show **min-max intervals** or **confidence intervals** for the output values. Under **Tools > Advanced Settings > Settings** you can select ☒ **Min-Max intervals with quartiles** if the *ExperimentManager* shows the 25 percent, the 50 percent, and the 75 percent quartiles for the min-max intervals.

This type of display is called a box plot. It shows the distribution of the observations of an output value of an experiment.

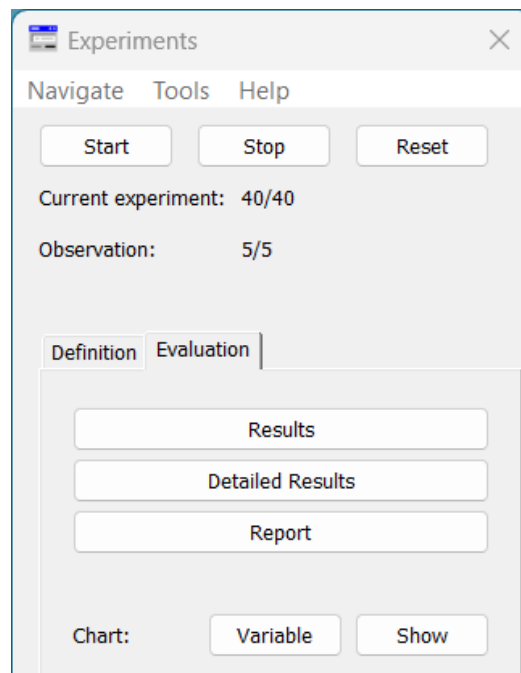


See also [View Output Values of Interest](#) in the *Object Reference* part of the *Plant Simulation Help*

View the Results in a HtmlReport

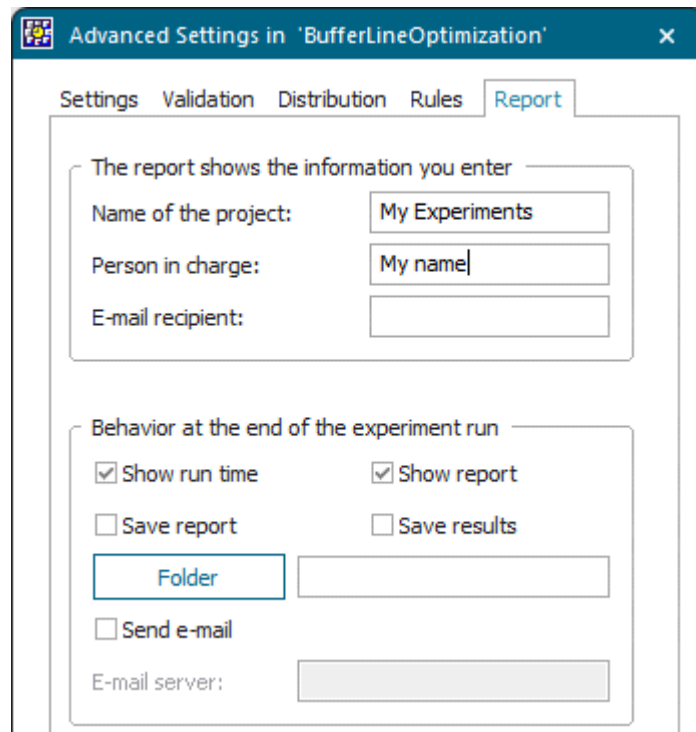
In this step we view the results of the simulation run in a *HtmlReport*.

To show the results of the simulation experiment as a [HtmlReport](#), click **Report**.



You can select what the report shows under **Tools > Advanced Settings > Report**. You can select to automatically show the report at the end of the simulation study and to save the report. You can also select if the *ExperimentManager* saves the data of the table *Results* to an Excel file. This way you can publish the results of a study in a compact format. If you want to save the report or the Excel file to a folder, click **Folder** and select the target folder.

To add a *Chart* which you inserted into your model to the *Report*, drag the *Chart* over the icon of the *ExperimentManager* and drop it there.



The screenshot shows a software window titled "Advanced Settings in 'BufferLineOptimization'". It has five tabs: "Settings", "Validation", "Distribution", "Rules", and "Report". The "Report" tab is selected. The window contains two main sections. The first section, titled "The report shows the information you enter", has three input fields: "Name of the project:" with the value "My Experiments", "Person in charge:" with the value "My name", and "E-mail recipient:" which is empty. The second section, titled "Behavior at the end of the experiment run", contains four checkboxes: "Show run time" (checked), "Show report" (checked), "Save report" (unchecked), and "Save results" (unchecked). Below these are two input fields: "Folder" (with a button icon) and an empty field. At the bottom, there is a checkbox for "Send e-mail" (unchecked) and an "E-mail server:" field which is disabled (grayed out).

See also

[Show the Results in a Report](#) in the *Reference* part of the *Help*

Refining the Settings

Refine the Settings

[Run a Simple Simulation Study](#) describes the most basic way of running experiments with the *ExperimentManager*. Here you learn how to refine the settings when you [Set Static Parameters](#) and when you [Set Dynamic Parameters](#).

Setting Static Parameters

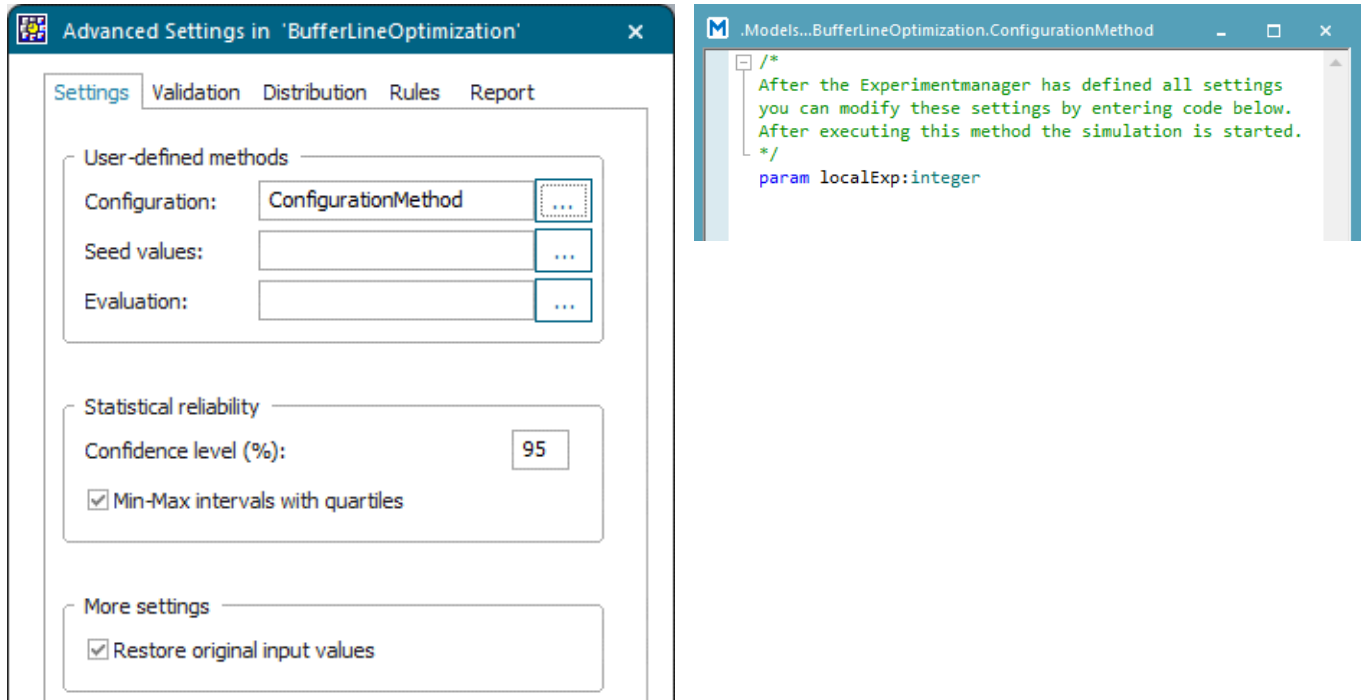
Set Static Parameters

Static parameters are specific values for each experiment for each parameter. If you want to, you can also [Modify Settings in the Configuration Method](#).

Modify Settings in the Configuration Method

Under **Tools > Advanced Settings > Settings** you can configure the settings of your simulation model with a *SimTalk* method.

Click the  button, select a *Method* in the dialog **Select Object** and click **Apply**. *Plant Simulation* asks if you would like to format the *Method*. The parameter of data type *integer* designates the number of the experiment whose additional settings you can program. You can use all commands which *SimTalk* provides. The source code you enter into this *Method* modifies the settings for the experiment.



[Back to Set Static Parameters](#)

Setting Dynamic Parameters

Set Dynamic Parameters

Under **Tools > Advanced Settings > Rules** you can define *dynamic parameters*. For this type of parameterization the results of previous experiments produce the values for the input values of the current experiment.

A **rule** is a logic expression that consists of a condition and an action. As soon as the condition of a rule is met, the associated action of the rule is taken.

First enter the number of experiments you want the *ExperimentManager* to execute.

The *ExperimentManager* checks if the conditions you entered as the **table condition** and as the **method condition** on the tab do apply. It then takes the action(s) which you typed into the **table action** and programmed in the **method action**.

Procedure

1. To use dynamically changing parameters, select the check box **use rules**.
2. Select one of the built-in rules from the drop-down list **Select rule**, [Create a Rule of Your Own](#).
3. Select the check box **Rule is active**, to activate the rule you selected. This rule will then be executed before the simulation experiment.
4. If you want to, you can enter the number of the experiment(s) for which the rule applies.
5. Enter a **Priority**. The greater the number you enter, the higher the priority will be: A rule with the priority 10 is executed before a rule with the priority 1.
6. To use the rule which you selected for the first experiment, select the check box **Use for first simulation run**. When you do not select it, the *ExperimentManager* only uses the rule from the second experiment onward.

Back to [Refine the Settings](#)

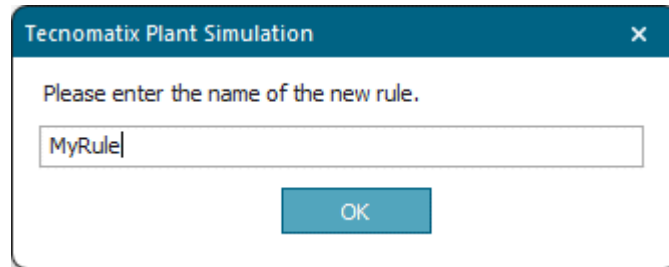
Create a Rule of Your Own

In this step we create a rule of our own.

Proceed as follows:

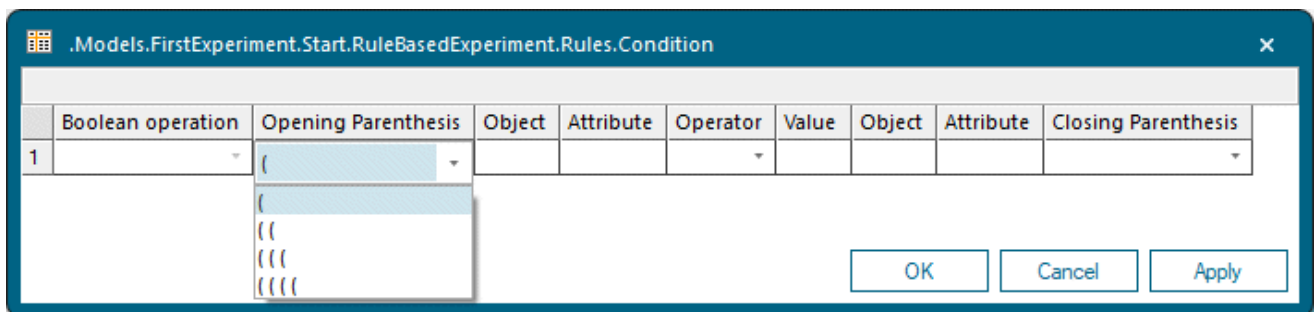
Procedure

1. Click the button **Create New Rule** and enter a name for the rule into the dialog that opens.

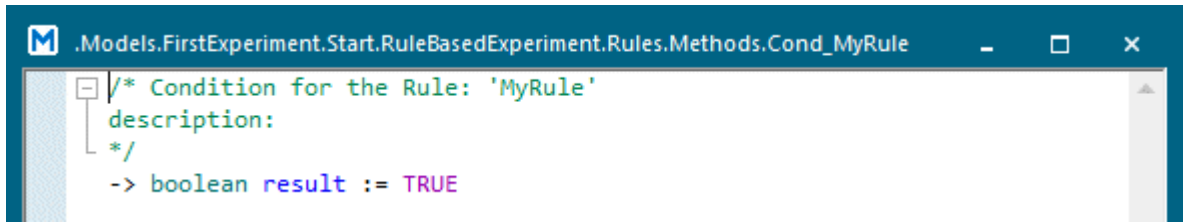


The *Plant Simulation* conventions for the **Name** of an object do apply.

2. Click **Table Condition**. Enter your values into the table that opens.



- Click into the cell and select the level of the **Opening Parenthesis**. *Plant Simulation* evaluates expressions in parenthesis from the inside to the outside, i.e., it evaluates the contents of the innermost expression in parenthesis first and then proceeds toward the outside one by one.
 - Enter the name of the **Object** whose attribute value you want to compare to another object.
 - Enter the name of an **Attribute** of this object.
 - Click into the cell and select an **Operator** that compares the attribute values: < (less than), > (greater than), = (equal to), ~= (about equal), or != (unequal to).
 - Either enter the **Value** that the operator compares or the name of the **Object** whose attribute value is compared to the object you entered above.
 - Enter the name of an **Attribute** of this object.
 - Click into the cell and select the level of the **Closing Parenthesis**.
 - Click into the cell **Boolean Operation** and select an **operator**, AND or OR that connects the boolean values that result from evaluating the logical expression you defined in a row of the query table.
3. If you want to, you can also test a condition in a *Method*. For this, click **Method Condition** and enter your source code into the *Method* that opens.



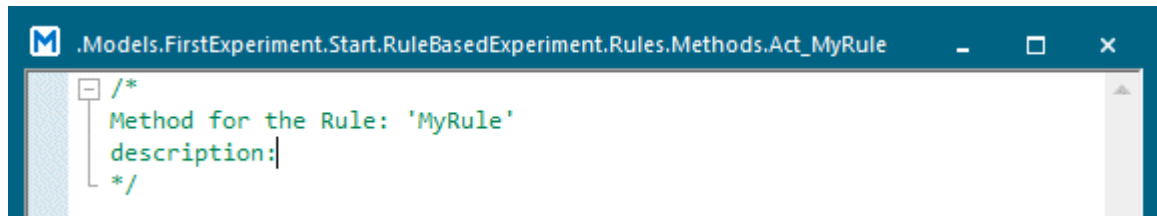
4. Click **Table Action**. Enter your values into the table that opens:

100

	Object	Attribute	Operator	Value
1	Delivery	SetupAfterNumParts	=	100

OK Cancel Apply

- Enter the name of the **object** whose attribute value you want to manipulate.
 - Enter the name of an **attribute** of this object.
 - Click into the cell and select an **operator** that manipulates the attribute value.
 - Enter the **value** that the operator adds, subtracts or equates to the value of the attribute you entered above.
5. If you want to, you can also program a *Method* that executes an action. To do so, click **Method Action** and enter your source code into the *Method* that opens.

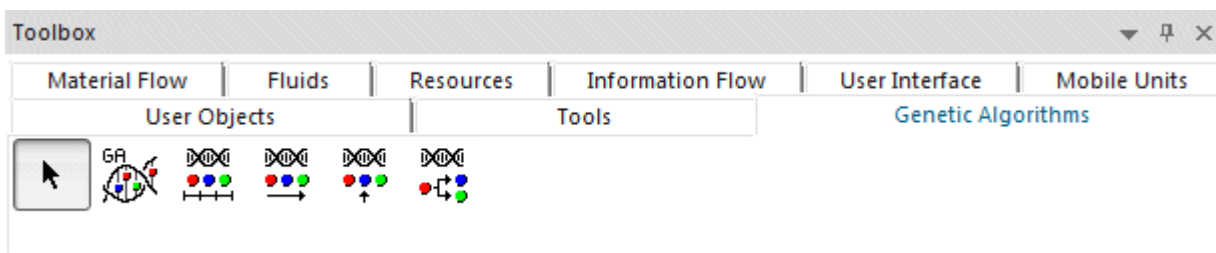


Back to [Set Dynamic Parameters](#)

Optimizing Models with Genetic Algorithms

Optimize Models with Genetic Algorithms

When your optimization task has a large number of different solution variants, we recommend to use *Genetic Algorithms*.



The **GA Wizard** integrates genetic algorithms into an existing simulation model. You can use it for optimizations in which the evaluation is based on a simulation run as well as for optimizations in which the evaluation is based on calculations in *Methods*.

Note

Your *Plant Simulation* license determines how you can use the *GA Wizard*.

Genetic Algorithms are stochastic optimization procedures that in most cases only produce an approximate solution. For the most common practical applications these approximate solutions do suffice. *Genetic algorithms* manage several solution proposals during the optimization. According to evolution in biology these solution proposals are called **individuals**. The individuals managed in a single optimization step are combined into a **generation**. The quality of a solution proposal is evaluated through a numerical value, the so-called **fitness value**.

Solutions generated by *Genetic Algorithms* are passed to the simulation model, which in turn will be configured accordingly. *Plant Simulation* starts one or several simulation runs for each solution attempt. At the end of a simulation run the simulation model is going to pass the resulting fitness value back to the object *GA Wizard*.

When your evaluation takes place via a model containing random components, such as failures of machines, then you have to execute several simulation runs for each newly created individual. The fitness value is called observation. The genetic algorithm uses the mean value of all observations of the fitness values of an individual. Enter the number of observations per individual on the tab **Define** of the *GA Wizard*. For a deterministic simulation, you will use a single observation per individual.

When the genetic algorithm creates individuals defining the same parameterization, the *GA Wizard* recognizes this and uses the fitness values of the already evaluated individuals. The optimization run thus does not waste additional time for multiple evaluations of an individual.

Compare the sample models: Click the **Window** ribbon tab, click **Start Page > Getting Started > Example Models > Small Examples**. Then, select the respective **Category**, the **Topic**, and the **Example** in the dialog **Examples Collection**, and click **Open Model**.

Back to [Run Simulation Experiments](#)

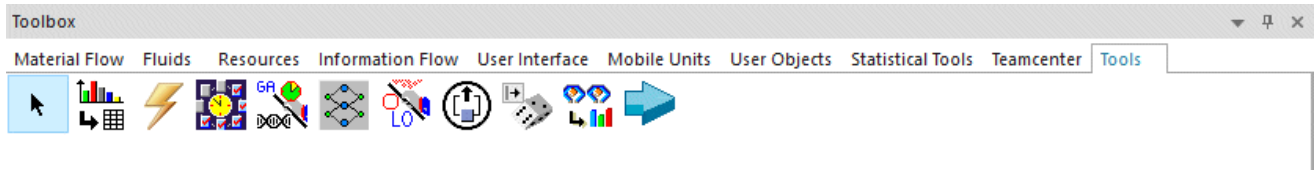
Configure the GAWizard

In this step we configure the *GAWizard*.

Proceed as follows to add the *GAWizard*:

Procedure

1. Click the toolbar **Tools** and insert the *GAWizard* into your simulation model. Double-click the icon to open the dialog of the *GAWizard*.

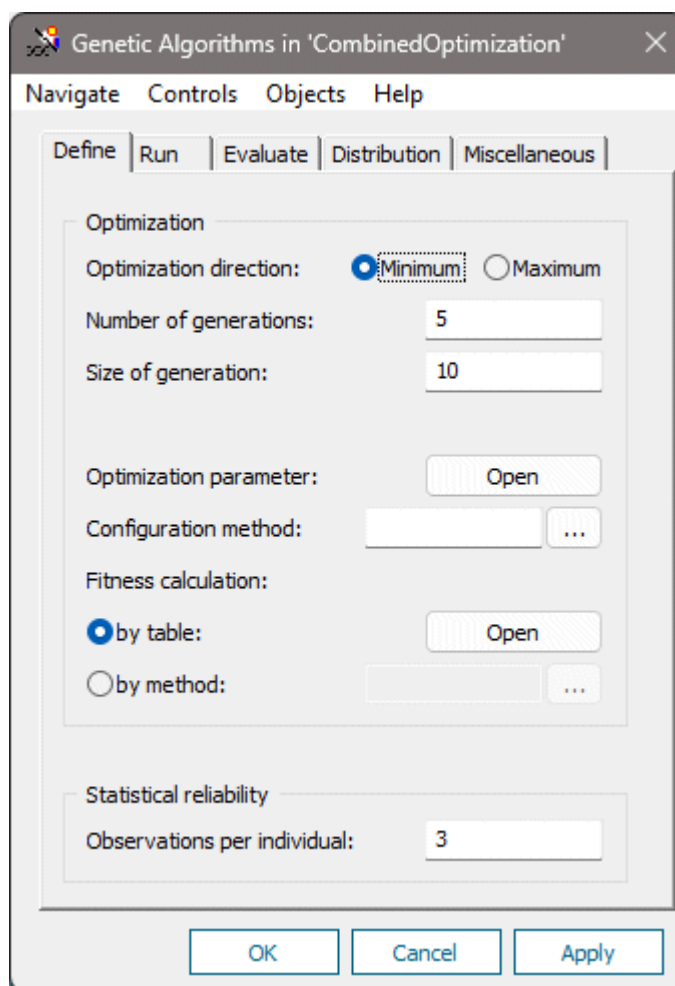


You can define the optimization problem with the *GAWizard* by using:

- Drag-and-Drop: Hold down **Shift** and drag the tables, the sequence of which you would like to optimize onto the *GAWizard* and drop it there. When you would like to optimize an attribute of an object, hold down **Shift** and drag this object onto the *GAWizard*, drop it and select the attribute. Then, you can change the value range of the parameters to be optimized into the table *ProblemDefinition*. To open this table, click the button **Open** next to **Definition** of the optimization parameter on the tab **Define**.

Drag the object, whose attributes and methods you would like to use for evaluating the created solutions, onto the *GAWizard* and drop them there. Then, select the attributes and methods of the object. To weight these values differently, click **Open** next to **Fitness calculation by table**.

- If you use the built-in *GA objects* for describing your optimization problem, first define a **configuration method** on the **Tab Define** of the *GAWizard*. Then hold down **Shift** and drag the *GA objects* (*GASelection* and/or *GASequence* and/or *GARangeAllocation* and/or *GASetAllocation*) onto the *Wizard* and click **Apply**. This registers all *GA objects* with the *GAWizard*. Then edit the **configuration method**. The attribute *individual* designates the table of the individual that is to be evaluated and that the genetic algorithm created. The first line of the source code shows how you can access the chromosomes of the individual. With the configuration method you can transmit all properties of the chromosomes to your simulation model. The result of the method is true when a parametrization can be executed with the values of the individual. When the parametrization fails, the respective individual will be assigned a bad fitness value, a so-called penalty value. The penalty value for unrealizable individuals is the attribute *penaltyValue* of the *GAWizard*. To change the value of this attribute, select the *GAWizard* in the *Frame*, into which you inserted it, and press the **F8** key.



2. Select the direction of the optimization: **Minimum** or **Maximum**. Enter the **Size of generation** and the **Number of generations** on the tab **Define**.
3. You can determine the fitness value with the *GAWizard* by using:
 - The table **Fitness**. Select the radio button **Fitness calculation by table** on the tab **Define** and click **Open**. This opens the table **Fitness**. Enter the names of the values into the cells of the column **Target value**. When the fitness value is determined by several target values, enter their weighting into the cells of the column **Weighting**.
 - You can let a method compute the fitness. To do so, select the respective radio button, insert a *Method*, and click **Apply**. If you want to, the *Wizard* can prepare a *Method* for computing the fitness. The return value is the fitness value and has to have a numeric data type, such as *real* or *time* for example
4. Click **Reset** and **Start** on the tab **Run** to start the optimization.

You can stop the optimization run after the individuals of the active generation have been completely evaluated and then modify the settings and the parameters of the object *GAWizard*. During the simulation run, you can click **Stop** on the tab **Run**. As long as *Plant Simulation* is still evaluating the individuals of the active generation the button shows **Wait**. When the button shows **Start** again you can click it to continue the optimization run for creating new generations.

After *Plant Simulation* is finished with the optimization run, it shows a dialog with the time the optimization run took. The wizard transfers the input values for the best solution it found into the simulation model. You can now analyze how the optimized model behaves.

To start additional optimization runs, click **Reset** and **Start** on the tab **Run** or click the icon of the *GAWizard* with the right mouse button and select **Reset** or **Start and Stop** on the context menu.

Note

The number of required simulation runs for an optimization can become quite large value.

For a stochastic simulation the individuals should be evaluated by several simulation runs. You can enter the number of simulation runs for evaluating an individual on the tab **Define > Observations per individual**. The number of individuals to be evaluated results from the **Number of generations** and from the **Size of generation**. You can enter both values on the tab **Define** as well.

In the first generation *Plant Simulation* evaluates the number of generations which you entered as the **Size of generation**. In each of the following generations it has to evaluate twice as many individuals. When, for example, you use 3 observations for evaluating an individual and create 10 generations with a frequently used **Size of generation** of 30, *Plant Simulation* has to execute 1710 simulation runs ($3 * (30 + 2 * 30 * 9)$).

The number of simulation runs, which the *GAWizard* executes, results from this formula:

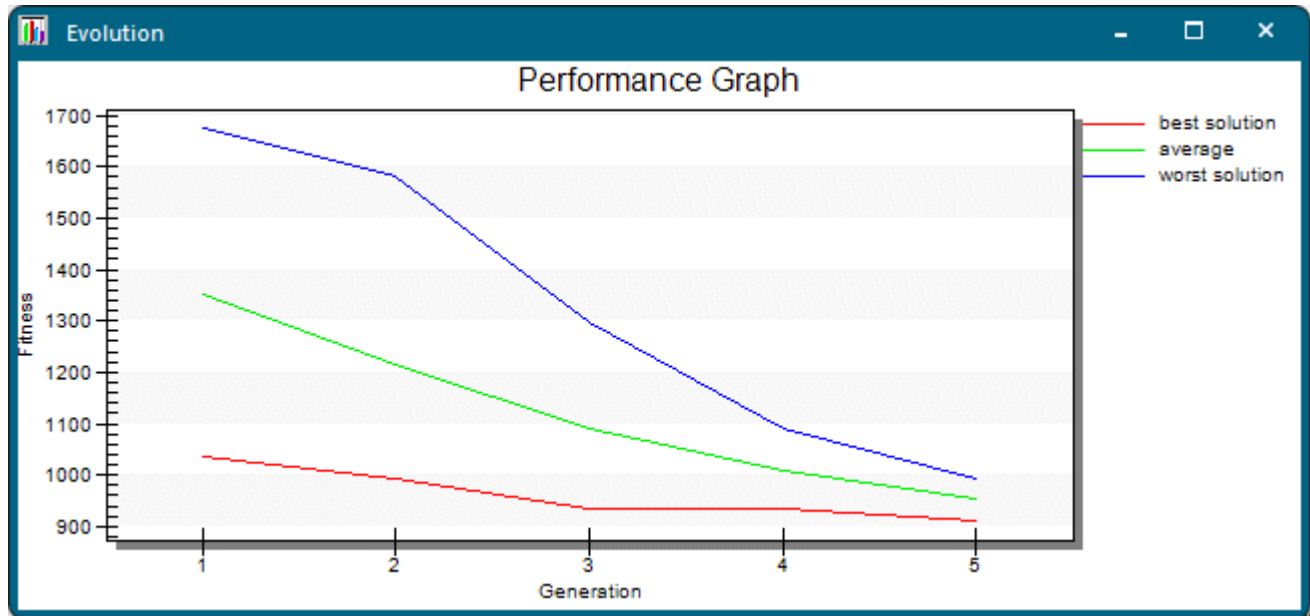
Number of simulation runs = observations per individual * (generation size + 2 * generation size * (number of generations - 1))

The individuals created at a generation change of the **Genetic Algorithm** might already have been created and evaluated in a previous generation. If the number of individuals to be generated is not greater than 50,000 the *GAWizard* recognizes these individuals and applies the fitness values. Such individuals have the same ID in the table **Statistics of Fitness Values** on the tab **Evaluate**. The attribute *lastGeneration4DuplicatesCheck* of the *GAWizard* describes the last generation up to which the already evaluated individuals will be checked. If you use the default value -1, the *GAWizard* runs this check for all generations.

Analyze the Results of the Optimization

On the tab **Run** you can observe the optimization process. It shows the currently best fitness value at any time. Click **Show Evolution** on the tab **Evaluate** to open a diagram displaying the best, the average and the worst fitness values of each generation.

You can, for example, detect, if additional optimization runs with the active settings produce better results or not.



Click **Show** in the group **HTML Report** on the tab **Evaluate** to show the results and the course of the optimization in a *Report*. When a very great number of individuals have been created, we recommend to clear the check box **Show detailed HTML report**.

You can, for example, detect, if the genetic algorithm is searching for the best results on the level of almost identical fitness values or if the fitness values of the offspring is still fluctuating quite a bit. For a stochastic simulation the chart shows Min-Max intervals of the fitness values of an individual an. When the fitness values fluctuate greatly, we recommend to check if an optimization of your model makes sense or not. To analyze the statistics table of the fitness values, click the respective buttons. Column 7 of the statistics table shows if individuals were created multiple times but evaluated once only.

Click **Show** next to **Data** of the report on the tab **Evaluate** to show the results and the history of the optimization. When *Plant Simulation* created a large number of individuals, we recommend to not select the check box **Show detailed report**.

Note

You can use the attribute *DecimalSeparator* of data type *string* to set the values "." (period) or "," (comma) as the decimal separator. The results tables then use this decimal separator, compare **Text File Format > Decimal Separator**.

For the Advanced User

Here we demonstrate techniques for the advanced user.

To start several optimizations one after the other with a BATCH file, compare [Specifying Start Options](#), *Plant Simulation* may not open any modal dialogs, which require user actions.